

ACR22113

## 4-Bromophenyl isocyanate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 异氰酸-4-溴苯酯  
4-Bromophenyl isocyanate

**Cat No. :** 221130000; 221130100  
**Synonyms** p-Bromophenyl isocyanate; 1-Bromo-4-isocyanatobenzene.  
**CAS No** 2493-02-9  
**Molecular Formula** C7 H4 Br N O

**Supplier** **UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
White

**Odor**  
pungent

#### Emergency Overview

Causes skin irritation. Causes serious eye irritation. May cause respiratory irritation. May be harmful if swallowed. Harmful in contact with skin. May cause an allergic skin reaction. Harmful if inhaled. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Moisture sensitive. Lachrymator (substance which increases the flow of tears).

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Acute Oral Toxicity                                | Category 5 |
| Acute Dermal Toxicity                              | Category 4 |
| Acute Inhalation Toxicity - Dusts and Mists        | Category 4 |
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Respiratory Sensitization                          | Category 1 |
| Skin Sensitization                                 | Category 1 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Label Elements

## 4-Bromophenyl isocyanate



## Signal Word

Danger

## Hazard Statements

H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H303 - May be harmful if swallowed  
H317 - May cause an allergic skin reaction  
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled  
H312 + H332 - Harmful in contact with skin or if inhaled

## Precautionary Statements

## Prevention

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P284 - In case of inadequate ventilation wear respiratory protection

## Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P342 + P311 - If experiencing respiratory symptoms: Call a POISON CENTER or doctor  
P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

None identified.

## Health Hazards

Causes skin irritation. Causes serious eye irritation. May cause respiratory irritation. May be harmful if swallowed. Harmful in contact with skin. May cause an allergic skin reaction. Harmful if inhaled. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Lachrymator (substance which increases the flow of tears).

## Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Other Hazards

Lachrymator (substance which increases the flow of tears)  
This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component                      | CAS No    | Weight % |
|--------------------------------|-----------|----------|
| Benzene, 1-bromo-4-isocyanato- | 2493-02-9 | 99       |

## SECTION 4. FIRST AID MEASURES

## 4-Bromophenyl isocyanate

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable. May cause allergic skin reaction. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Carbon dioxide (CO<sub>2</sub>). Dry chemical.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Avoid dust formation. Ensure adequate ventilation.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE**

## 4-Bromophenyl isocyanate

**Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage**

Keep in a dry place. Keep container tightly closed. Keep refrigerated.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component                         | ACGIH TLV | OSHA PEL | NIOSH | The United Kingdom                                                                     | European Union |
|-----------------------------------|-----------|----------|-------|----------------------------------------------------------------------------------------|----------------|
| Benzene,<br>1-bromo-4-isocyanato- |           |          |       | STEL: 0.07 mg/m <sup>3</sup> 15 min<br>TWA: 0.02 mg/m <sup>3</sup> 8 hr<br>Resp. Sens. |                |

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-----------------------------------------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

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**Recommended Filter type:** Particulates filter conforming to EN 143**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                                |                               |                                 |
|------------------------------------------------|-------------------------------|---------------------------------|
| <b>Appearance</b>                              | White                         |                                 |
| <b>Physical State</b>                          | Solid                         |                                 |
| <b>Odor</b>                                    | pungent                       |                                 |
| <b>Odor Threshold</b>                          | No data available             |                                 |
| <b>pH</b>                                      | No information available      |                                 |
| <b>Melting Point/Range</b>                     | 42 - 44 °C / 102.2 - 107.6 °F |                                 |
| <b>Softening Point</b>                         | No data available             |                                 |
| <b>Boiling Point/Range</b>                     | 158 °C / 316.4 °F             | @ 14 mmHg                       |
| <b>Flash Point</b>                             | 109 °C / 228.2 °F             | <b>Method -</b> CC (closed cup) |
| <b>Evaporation Rate</b>                        | Not applicable                | Solid                           |
| <b>Flammability (solid,gas)</b>                | No information available      |                                 |
| <b>Explosion Limits</b>                        | No data available             |                                 |
| <b>Vapor Pressure</b>                          | No data available             |                                 |
| <b>Vapor Density</b>                           | Not applicable                | Solid                           |
| <b>Specific Gravity / Density</b>              | No data available             |                                 |
| <b>Bulk Density</b>                            | No data available             |                                 |
| <b>Water Solubility</b>                        | No information available      |                                 |
| <b>Solubility in other solvents</b>            | No information available      |                                 |
| <b>Partition Coefficient (n-octanol/water)</b> |                               |                                 |
| <b>Autoignition Temperature</b>                | No data available             |                                 |
| <b>Decomposition Temperature</b>               | No data available             |                                 |
| <b>Viscosity</b>                               | Not applicable                | Solid                           |
| <b>Explosive Properties</b>                    | No information available      |                                 |
| <b>Oxidizing Properties</b>                    | No information available      |                                 |
| <b>Molecular Formula</b>                       | C7 H4 Br N O                  |                                 |
| <b>Molecular Weight</b>                        | 198.02                        |                                 |

**SECTION 10. STABILITY AND REACTIVITY**

|                                         |                                                                                                                                        |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Moisture sensitive.                                                                                                                    |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                          |
| <b>Hazardous Polymerization</b>         | No information available.                                                                                                              |
| <b>Conditions to Avoid</b>              | Heat, flames and sparks. Incompatible products. Exposure to moist air or water.                                                        |
| <b>Materials to avoid</b>               | Acids. Water. Strong oxidizing agents. Strong bases. Alcohols. Amines.                                                                 |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Hydrogen halides. Hydrogen cyanide (hydrocyanic acid). |

**SECTION 11. TOXICOLOGICAL INFORMATION**

## 4-Bromophenyl isocyanate

## Product Information

## (a) acute toxicity;

| Component                      | LD50 Oral                  | LD50 Dermal | LC50 Inhalation |
|--------------------------------|----------------------------|-------------|-----------------|
| Benzene, 1-bromo-4-isocyanato- | LD50: > 3200 mg/kg ( Rat ) |             |                 |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

## (d) respiratory or skin sensitization;

Respiratory

Category 1

Skin

Category 1

No information available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; Category 3

Results / Target organs

Respiratory system

(i) STOT-repeated exposure; No data available

Target Organs

No information available.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

Persistence and Degradability No information available

Bioaccumulative Potential No information available

Mobility in soil No information available

SAFETY DATA SHEET  
4-Bromophenyl isocyanate

|                                 |                                                                           |
|---------------------------------|---------------------------------------------------------------------------|
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance            |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance            |

SECTION 13. DISPOSAL CONSIDERATIONS

|                                     |                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point.                                                                                              |
| Other Information                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.                                          |

SECTION 14. TRANSPORT INFORMATION

|                              |                                  |
|------------------------------|----------------------------------|
| Road and Rail Transport      | Not Regulated                    |
| IMDG/IMO                     | Not regulated                    |
| IATA                         |                                  |
| UN-No                        | UN3335                           |
| Proper Shipping Name         | AVIATION REGULATED SOLID, N.O.S. |
| Hazard Class                 | 9                                |
| Packing Group                | III                              |
| Special Precautions for User | No special precautions required  |

SECTION 15. REGULATORY INFORMATION

|                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| International Inventories                                                                                                                                                 |
| X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL). |

| Component                      | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|--------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|------|
| Benzene, 1-bromo-4-isocyanato- | X                                                   | -                                       | X    | -     | 219-662-9 | X    | -   | -     | -    |      | -    | -    |

National Regulations

SECTION 16. OTHER INFORMATION

|                  |                 |
|------------------|-----------------|
| Revision Date    | 07-Apr-2024     |
| Revision Summary | Not applicable. |

## 4-Bromophenyl isocyanate

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**